Curriculum Council
Government of **Western Australia**

WACE Examination, Sample 2008

ENGINEERING STUDIES

Section 2, Materials, Structures and Mechanical Systems

STAGE 2

Provisional Marking Guidelines

TRIM: 2008/17466

Section 2

Part A: Multiple Choice Questions

1	A
2	B
3	D
4	C
5	D
6	B
7	B
8	C
9	C
10	A
	[1 mark each]

Part B: Written (Stage 2)

Q1

(a) Worm does 2000rpm;
40 revolutions of worm give one turn of the 80/40 wheel; [2]
2000/40 = 50rpm [2]
120 wheel will turn at 80/120 x 50rpm = 33.33rpm [2]
[Total 6]

(b) Circumference of drum = πd = 3.142 x 0.2 = 0.628m [2]
Drum turns at 33.33rpm; in 50 seconds turns 33.33 x 50/60 = 27.78 turns [2]
Distance is therefore 27.78 x 0.628 = 17.44m [2]
[Total 6]

(c) Many different ways of expressing torque, e.g. "Rotational force" or "Turning force" analogous to Force in non-rotating systems; Force times lever arm; Force times perpendicular distance from the pivot to the line of action of the force.
[Total 2]

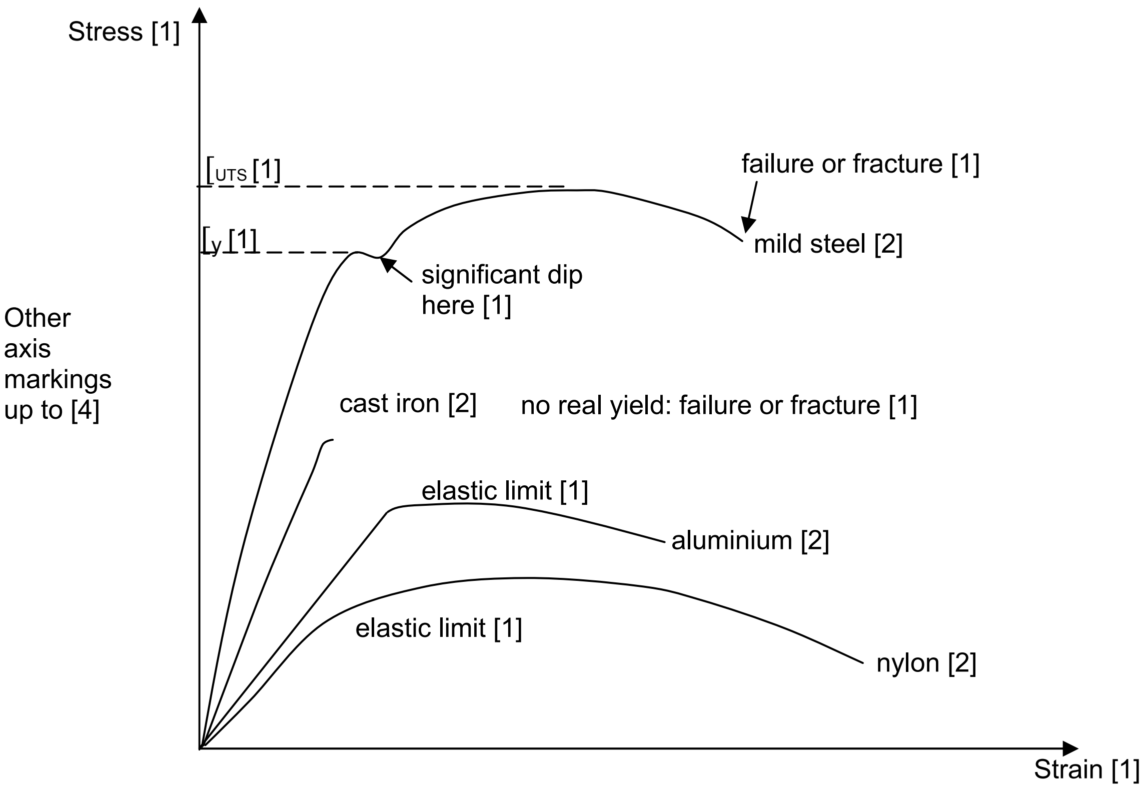
(d) T = force x radius; 600 x 0.1 = 60 Nm
[Total 2]

(e) Torque from part (d) is output torque; input torque from graph is 11.5N Answer between 11 and 12 is OK.
[Total 4]

Q2

(a) Tensile test. Specimen is help at each end [1] in the jaws of a tensometer, which applies a steadily increasing load [1] to the specimen while recording load and extension [1]. Load and extension are used to calculate stress and strain.[1]
[Total 4]

(b)

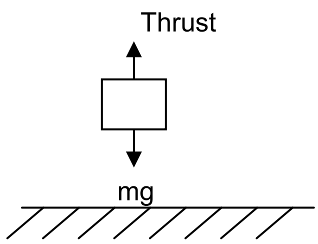


[Total 20]

(c) $0.2m^2$; 40kN applied, stress = F/A [1]
= 40 000/0.2 = 200 000 N/m^2 (or Pa) or 200 kNm^2 [1]
E = stress/strain = [1]
E = 200 000/0.003 = 66.67MPa (or MN/m^2) [1]

Q3

(a) See figure below.



Arrows in right directions [2]
Downward force identified as mg [1]

[Total 3]

(b) Net force = Thrust – (mg) = 16000-(4550x1.62) = 8629N [2]
F=ma; a=F/m = 8629/4550 = 1.896 = 1.90 m/s^2 [2]
After 10 seconds, v=u+at (u=0) v = 1.896x10 = 19.0m/s [2]
$S=ut+½at^2$; (u=0) s = $½at^2$ = 1.896 x 100/2 = 94.8m [2]
[1 mark for suitable rounding of answers i.e. to no more than 2 dp]
[Total 9]

(c) v=u+at again, v = 1.896 x (7x60 + 14) = 822.9 m/s [2]
Acceleration will not stay constant because the mass is decreasing as the fuel is consumed. The acceleration will therefore increase as time passes, and the final velocity will be much higher.[2]
[Total 4]

(d) Work done = force x distance; [1]
16kN x 308km; 16 000 x 308 000
4.93 GJ [1 plus 1 for sensible units]
[Total 3]

(e) K.E = ½mv2 [1]
Change in K.E = $½m(V_1^2 - V_2^2)$
= $5568/2 \times (11032^2 – 9.5^2)$ [2]
= 3.39×10^{11} J [1] = 339 GJ [1 for sensible units]
[Total 5]

Q4

(a) From data book, $\rho_{brass} = 8740\ kg/m^3$ [1]
x-sectional area A = $3/2\ \sqrt{3}\ a^2$ where a = 6mm = 0.006m
A = $9.353 \times 10^{-5}\ m^2$ [1]
A x ρ = ω mass per unit length i.e. kg/m
Mass per unit length = $9.353 \times 10^{-5} \times 8740$ = 0.817 kg/m [2]
Therefore ω = 9.80 x 0.817 = 8.0 N/m [2]
[Total 6]

(b) ω = 8.0 N/m from part (a)
L = 0.745 m (given)
E = 110 kN/mm^2 from data book; $=110 \times 10^9\ N/m^2$ [2]
I = $7.015 \times 10^{-10}\ m^4$ (given)
So $\delta = 8.0 \times 0.745^4/(8.\ 110 \times 10^9.\ 7.015 \times 10^{-10}) = 4.00 \times 10^{-3}$m = 4mm [2 for answer, 1 for consistent units, 1 for suitable rounding]
[Total 6]

(c) Reduce length [1] - deflection proportional to L^4 so small reduction will have large effect. [1]
Change material [1] – higher Young's Modulus will give smaller deflection. [1]
Increase bar diameter [1] – will increase I and therefore reduce deflection. [1]
[Total 6]